Coding & Solution

Date	12 July 2026
Team ID	6a2fbf036b1ed0bd8fa789aa
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/sb-dr-01/ai-debt-relief-platform
Programming Language(s)	Python, JavaScript (TypeScript optional for frontend)
Framework(s) Used	FastAPI (backend), React.js + Vite (frontend), SQLAlchemy (ORM)
Key Features Implemented	- Borrower login/registration with JWT authentication - Loan detail input and management - Financial health engine (EMI ratio, monthly surplus, debt stress score) - AI-powered settlement prediction with rule-based fallback - Gemini AI-integrated negotiation letter generator - Financial dashboard with visual health tracking - AI negotiation history module
Pending / Incomplete Features	- Multi-language support for negotiation letters - Automated lender response tracking / status updates - Admin analytics dashboard for aggregated borrower insights - Migration from SQLite to PostgreSQL for production scale
Setup / Run Instructions	1. Clone the repository: git clone https://github.com/sb-dr-01/ai-debt-relief-platform 2. Backend: cd backend, create a virtual environment, pip install -r requirements.txt, set GEMINI_API_KEY in .env, then run uvicorn main:app --reload 3. Frontend: cd frontend, npm install, npm run dev 4. Access the app at http://localhost:5173 (frontend) with the API running at http://localhost:8000

Code Quality Checklist

S.No	Criteria	Status (Yes/No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

Core borrower-facing features (authentication, loan management, settlement prediction, and AI negotiation letter generation)areimplementedandfunctionalend-to-end.Remainingworkfocusesonproductionhardening(PostgreSQL migration, lender response tracking) and admin-facing analytics.